

Sabuj C. Bhowmick

Data Scientist | Machine Learning Researcher | Computational Biology

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Profile

MSc Data Analytics student at the University of Turku with an interdisciplinary background in statistics, mathematical sciences, public health, IoT engineering, machine learning, and computational research. Focused on reproducible and interpretable data science, rigorous model validation, bioinformatics, multi-omics integration, and research-oriented software development.

Education

MSc	University of Turku , Information and Communication Technology - Data Analytics	2025 – present
	- Graduate studies in machine learning, data analytics, statistical modeling, and data-intensive systems. - Focus on reproducible analysis, model validation, and research-oriented computing.	
BEng	Savonia University of Applied Sciences , Internet of Things	2024
	Engineering studies in connected systems, data acquisition, software development, and applied analytics.	
MSc	Norwegian University of Science and Technology , Mathematical Sciences	2019
	Graduate training in mathematical reasoning, quantitative modeling, and scientific problem solving.	
MPhil	University of Oslo , International Community Health	2015
	Research training in public health, epidemiological reasoning, and quantitative health research.	
MPS	University of Dhaka , Population Sciences	2013
MSc	University of Dhaka , Statistics	2011
	Advanced training in probability, statistical inference, regression, and applied data analysis.	
BSc	University of Dhaka , Statistics	2010

Experience

Savonia University of Applied Sciences , Project Worker - 5G to Boost Digitization	June 2023 – Aug 2023
Developed applied data-analysis and forecasting workflows within a 5G- and IoT-oriented digitalization project.	3 months
- Built solar-energy production analysis and forecasting workflows. - Compared statistical and machine learning models, including ARIMA, random forest, SVR, and XGBoost. - Integrated weather data and prepared reproducible technical reporting.	
Voimatel Oy / Savonia University of Applied Sciences , Intern - Starlink Feasibility Testing	Sept 2023 – Jan 2024
Evaluated satellite connectivity for remote operational scenarios and compared it with mobile-network alternatives.	5 months
- Measured latency, bandwidth, signal reliability, and field performance. - Documented equipment setup, test conditions, results, and practical limitations.	

- Contributed to structured evidence assessment and interdisciplinary scientific review.
- Applied methodological scrutiny to research interpretation and reporting.

Skills

Programming and research software: Python, R, SQL, Bash, Git, GitHub, Quarto, R Markdown, LaTeX

Machine learning: scikit-learn, PyTorch, XGBoost, random forest, SVM, regression, classification, cross-validation

Statistics: applied statistics, statistical inference, time-series analysis, biostatistics, model evaluation

Bioinformatics: Bioconductor, microbiome analysis, compositional data, multi-omics integration, Joint-RPCA

Engineering practices: reproducible pipelines, automated testing, data validation, version control, technical documentation

Projects

Joint-RPCA for Multi-omics Integration

Cross-language validation of R/Bioconductor and Python/Gemelli workflows for microbiome multi-omics data.

- Developed and tested consistent input filtering for multi-assay biological count data.
- Compared sample scores, feature loadings, explained variation, and cross-validation outputs.
- Produced reproducible Quarto workflows with fixed configurations and interoperable outputs.

Solar Energy Forecasting

Comparative forecasting with statistical and machine learning models.

- Integrated photovoltaic production and weather data.
- Compared ARIMA/ETS, random forest, SVR, and XGBoost using chronological validation.

Protein-Drug Affinity Prediction

Group-aware validation study for affinity prediction.

- Compared leave-one-observation, leave-one-protein, and leave-one-drug validation.
- Identified a substantial loss of generalization for previously unseen proteins.

Languages

Bangla: Native

English: Professional working proficiency

Finnish: Basic

Interests

Research interests: computational biology, bioinformatics, applied AI, biostatistics, scientific computing, reproducibility